Academic Writing Coach: Week 3 Check-in: Users and Data

Track: Writing Assistant
Domain: Academic Writing Support
Core HAI Principle: Transparency
Project Type: Interactive AI Writing Coach
Document Status: Week 3 Design Draft
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1. Project Overview

The Academic Writing Coach is an advanced conversational assistant designed specifically for university students who know how to draft essays, research papers, and term papers but frequently struggle with academic phrasing, citation standards, formal tone, and structural coherence. Unlike generic AI writing tools that immediately rewrite student text or generate ungrounded fabrications, the Academic Writing Coach acts as an interactive pedagogical consultant. Students submit draft excerpts along with specific writing queries and work collaboratively with the mentor to understand structural weaknesses and refine their prose independently.

The system combines two complementary interaction pathways: (1) **Conversational Mentoring**, where students discuss their arguments, explore stylistic alternatives, and request incremental hints, and (2) **Rule-Based Verification**, where students select specific style manual criteria (APA, MLA, IEEE) to validate in-text citations and bibliographic references.

Crucially, the system adheres strictly to the Human-AI Interaction principle of **Transparency**. Every feedback item, structural critique, and formatting correction is explicitly anchored to authoritative institutional style guides. When departmental guidelines conflict or stylistic choices remain open to instructor discretion, the system explicitly communicates its uncertainty rather than issuing arbitrary mandates.

2. Detailed Fictional User Personas

Note: The following personas are fictional design profiles created solely for prototyping and requirement analysis. They do not represent real individuals, and no real student data was utilized.

Persona 1: Youssef Ibrahim — Below-Typical Technical Level & Accessibility Need

Background: Youssef is a fictional first-year university student studying computer science at Benha University. While comfortable with mobile messaging apps, he has minimal experience interacting with AI writing tools. Dense academic style manuals and complex citation guidelines overwhelm him.

Technical Level: Below typical for the intended user group. **AI Experience:** Novice AI user unfamiliar with prompt engineering or verifying generative outputs. **Cultural Background:** Egyptian; native Arabic speaker preferring clear, jargon-free explanations.

Accessibility Need: Youssef experiences mild visual fatigue and contrast sensitivity. The interface must avoid communicating error severity or badge statuses through color alone, employing readable typography, explicit text labels, and recognizable icons.

Goals: Understand basic essay formatting rules; learn how to structure introductory paragraphs; fix citation errors without losing his original voice.

Pain Points: Unfamiliar pedagogical terminology; dense style guides; fear of accepting incorrect AI rewrites.

Expected System Support: Plain-language explanations, structured citation templates, incremental hint levels, and accessible high-contrast UI elements.

Persona 2: Mariam Adel — Typical Technical Level

Background: Mariam is a fictional second-year student who frequently utilizes digital platforms. She writes coherent paragraphs but struggles with proper in-text citation formatting (APA vs. IEEE) and maintaining a consistently formal academic tone.

Technical Level: Typical. **AI Experience:** Novice-to-moderate. Occasionally uses chatbots for brainstorming but rarely verifies citation authenticity.

Cultural Background: Egyptian; comfortable with bilingual academic environments.

Goals: Receive constructive feedback on research drafts; distinguish between mandatory formatting rules and optional stylistic preferences.

Pain Points: Lack of transparency regarding why certain sentences are flagged; opaque AI rewrites that alter core arguments.

Expected System Support: Rule-based citation verification, clear explanations distinguishing strict rules from stylistic suggestions, and easy feedback controls.

Persona 3: Ziad Khaled — Advanced Technical Level

Background: Ziad is a fictional third-year student comfortable with advanced literature reviews, research synthesis, and rigorous academic argumentation. He debugs complex logical flow independently but seeks granular rhetorical diagnostics.

Technical Level: Advanced relative to student body. **AI Experience:** Experienced AI user who routinely evaluates model reliability.

Cultural Background: Nigerian studying in an international university setting.

Goals: Access direct style manual references instantly; validate advanced structural synthesis; avoid repetitive explanations.

Pain Points: Generic writing tools offering superficial praise or hallucinating citation rules.

Expected System Support: Concise technical critiques, direct access to style manual citations, and explicit acknowledgement of rhetorical uncertainty.

3. Comprehensive Bias-Check Notes

The team conducted a systematic bias and inclusion review across the design artifacts:

- **3.1 Cultural and Linguistic Neutrality:** Fictional backgrounds reflect diverse student profiles, but technical and writing proficiency are specified independently. Culture is never linked to academic aptitude.
- **3.2 Accessibility-First Design:** Visual contrast needs (Youssef) are addressed without treating accessibility as a deficit, avoiding color-only status indicators.
- **3.3 Handling Uncertainty Without Bias:** The transparency framework ensures the system never enforces subjective stylistic norms as absolute universal rules, preventing biased grading.

4. Data Source Description

4.1 Proposed Data Source: The project utilizes a team-curated synthetic dataset (`academic_guidelines_dataset.json`) targeting 40 structured cases. It contains zero real student personal data or private drafts.

4.2 Dataset Schema: Each record includes: (1) Case ID, (2) Category, (3) Student Draft Snippet, (4) Observed Issue, (5) Expected Rule, (6) Style Manual Reference, (7) Uncertainty Fallback Notes, (8) Progressive Hints, (9) Verified Reference Feedback, and (10) Validation Tests.

4.3 Initial Dataset Categories

Category	Planned Count	Examples
Grammar & Syntax	10	Passive voice overuse, run-on sentences, tense inconsistency
Citation & Formatting	12	APA missing year, IEEE bracket format errors, malformed bibliography
Structural & Argumentative	12	Weak thesis statements, unsupported claims, poor paragraph transitions
Insufficient Context Cases	6	Ambiguous prompt excerpts requiring clarification before critique
Total	**40**	—

4.4 Data Quality and Validation: Cases are verified against official APA/MLA/IEEE style guides. Ambiguous prompts are explicitly marked to trigger clarifying questions rather than ungrounded fabrications.

5. Prompt Log — Appendix A

Entry A1 — ChatGPT: Core Interaction Flow (Transparency)

Tool: ChatGPT | *Purpose:* Develop core conversational mentoring flow focusing on Transparency.

Prompt: Design a core conversation flow for an AI Academic Writing Coach supporting Transparency by grounding all feedback in official style manuals.

Output Summary: Suggested a draft submission workflow paired with style guide citations.

Team Changes: Added explicit uncertainty fallback messaging when style guidelines are ambiguous.

Entry A2 — Google Gemini: Rule-Based Citation Verification

Tool: Google Gemini | *Purpose:* Explore alternative verification pathways for citation checking.

Output Summary: Proposed a structured rule-matching mode for APA and IEEE reference formatting.

Team Changes: Integrated rule-matching alongside conversational mentoring as dual interaction paths.

6. Week 3 Check-in Readiness Checklist

- Three fictional personas included across technical levels and AI familiarity.
- Accessibility needs and contrast requirements fully documented.

  Bias-check notes review cultural neutrality and uncertainty handling.
- Synthetic dataset (`academic_guidelines_dataset.json`) structured with 40 target cases.
- Prompt log entries (A1, A2) document exact prompts and team adaptations.